

EARLY-STAGE RESULTS

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses



Last week's key findings Quick recap.

1

We found each person's "beauty formula"

A set of 7 numbers showing how much each person cares about each emotion when judging beauty.

2

Beauty preference is personal at the perception level

When we replaced a person's own emotion scores with the average from other people viewing the same image, accuracy dropped from 0.72 → 0.33.

3

Hybrid pipeline beats Direct prediction at every data level

Biggest gain: +40% when we only have 10 images per person (0.153 vs 0.109 CCC).

This week: two analysis questions about the personal formula.

RQ3 Does the formula connect to personality?

Do users who score high in Openness also care more about intellectually challenging images?

Key Questions Explored:

- Test whether emotion weights connect to broad personality traits.
- Look for links between Openness and intellectually challenging images.
- Check whether negative emotion weights align with Emotional Stability.

RQ4 Does the formula stay the same across image types?

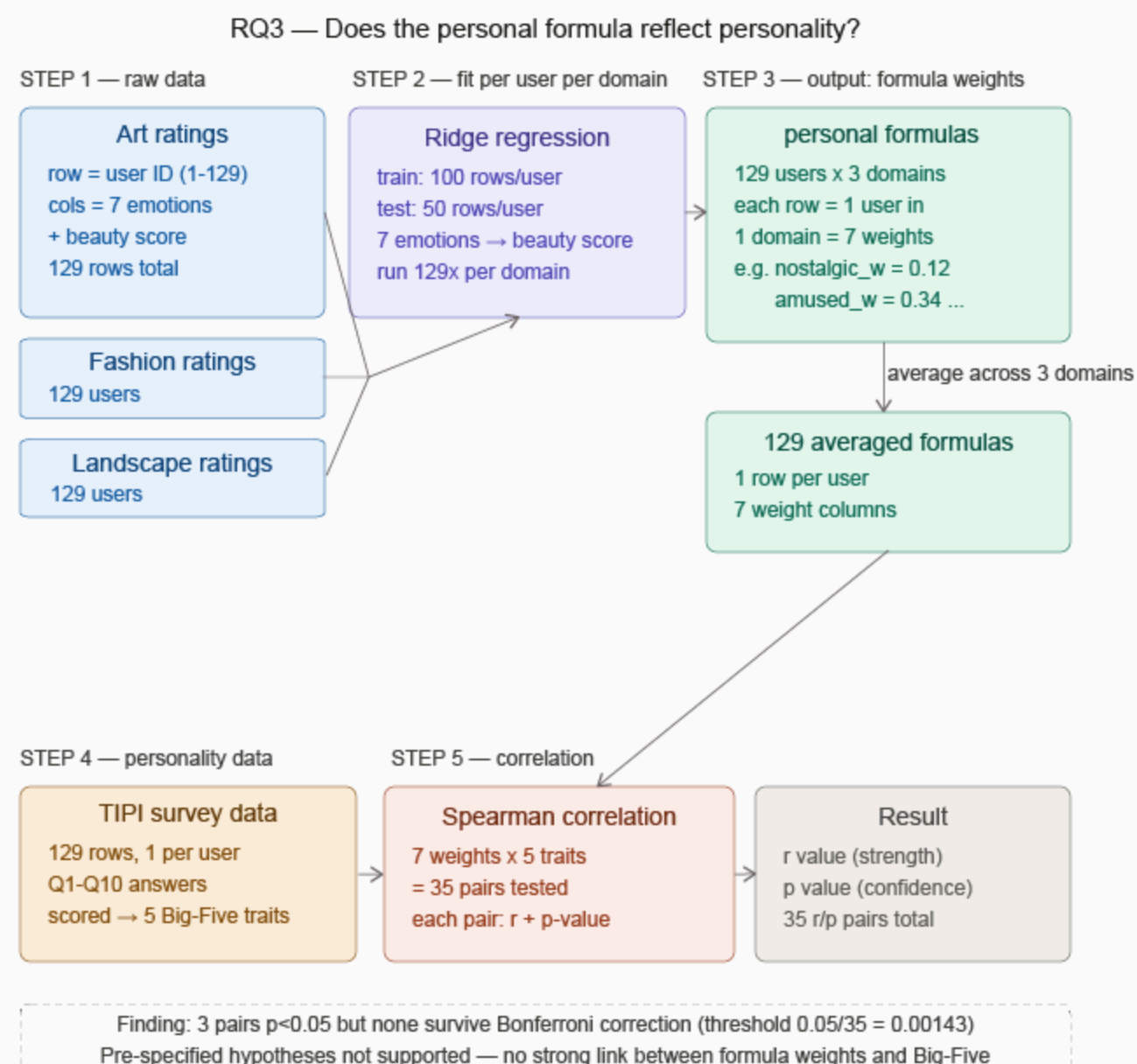
If someone cares a lot about nostalgia in art images, do they also care about it in fashion images?

Key Questions Explored:

- Check whether the personal formula is stable across Art, Fashion, and Landscape.
- Compare same-user consistency against a permutation baseline.
- Identify which emotions are most and least stable across domains.

RQ3 — How we tested it Pipeline & Design.

PIPELINE SETUP



1 Fit a personal formula per user per domain

Used real emotion scores, not predicted ones, plus beauty scores as input.

Ridge Regression produces **7 weight numbers** per person (e.g. nostalgic_w = 0.12).

Done for Art, Fashion, and Landscape, then averaged to 1 formula per user.

2 Calculate Big Five personality scores

XPASS-Vis already has TIPI survey answers for all 129 users.

Applied the standard Gosling et al. 2003 formula to get 5 trait scores per person.

3 Measure correlation between formula and personality

Use Spearman rank correlation for every emotion weight × every trait.

7 emotions × 5 traits = **35 pairs tested**.

RQ3 — What we found **Weak personality link.**

3 pairs appeared significant at $p < 0.05$

PAIR RELATIONSHIP	CORRELATION (R)	P-VALUE	SURVIVES BONFERRONI?
Impressed Weight × Conscientiousness	-0.18	0.041	No ($p > 0.0014$)
Motivated Weight × Openness	+0.18	0.037	No ($p > 0.0014$)
Sad Weight × Agreeableness	+0.20	0.024	No ($p > 0.0014$)

** But we tested 35 pairs — Bonferroni correction requires $\alpha = 0.0014$ ($0.05 / 35$).*

Hypothesis Test Count: 35

✗ **Pre-specified hypotheses not supported**

- **Openness × intellect weight:** $r = -0.10$, $p = 0.26$.
- **Emotional Stability × negative emotion weights:** $r = -0.01$, $p = 0.95$.

⚠ **Statistical Tempering**

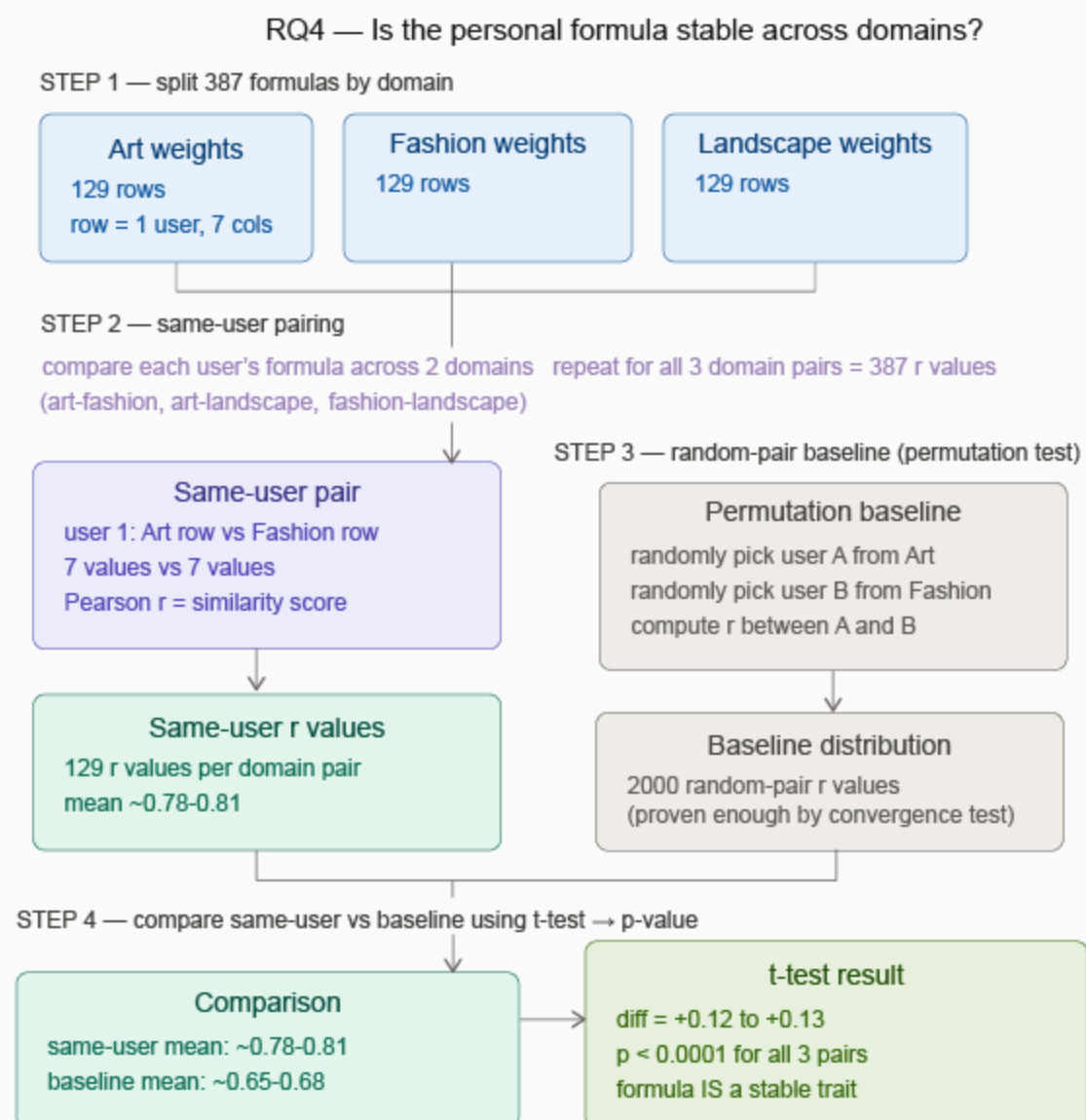
- **Small effects:** Correlations range around $r = 0.18$ to 0.20 .
- **False positives:** 1–2 false positives are expected by chance alone under 35 tests.

💡 **Honest Summary**

No strong evidence that the personal formula reflects Big Five personality. Personal preferences are more specific than broad traits can capture, consistent with Experiment 2 finding that 88% of personalization comes from perceiving emotions differently, not weighting them differently.

RQ4 — How we tested it Pipeline & Design.

PIPELINE SETUP



1 Fit a personal formula per user per domain

Starting from the formula fitting earlier, each user has 3 formula vectors: one from Art, one from Fashion, and one from Landscape. Each vector has 7 values.

2 Same-user correlation

For every user, calculate the Pearson correlation coefficient between their weight vectors for each pair of domains. A high correlation indicates a consistent personal aesthetic formula. Repeat for all 3 domain pairs × 129 users = 387 r values total.

3 Permutation baseline test

Randomly pick User A's Art formula and User B's Fashion formula (different people). Compute r between them and repeat 2000 times to build the random-pair distribution. Then compare whether same-user r is significantly higher than the baseline.

RQ4 — What we found

Stable personal signatures.

Domain pair | Same-user r | Random-pair r | Difference | p-value

Domain pair	Same-user r	Random-pair r	Difference	p-value
Art & Fashion	0.776	0.655	+0.120	<0.0001
Art & Landscape	0.786	0.654	+0.132	<0.0001
Fashion & Landscape	0.805	0.682	+0.124	<0.0001

* The same-user formula is significantly more consistent than random pairs across all 3 domain combinations.

Permutations Run: 2000



Emotion Consistency Rankings

😄 Amused (Most Consistent)

r = 0.57

🕒 Nostalgic (Least Consistent)

r = 0.14



Shared Baseline Context

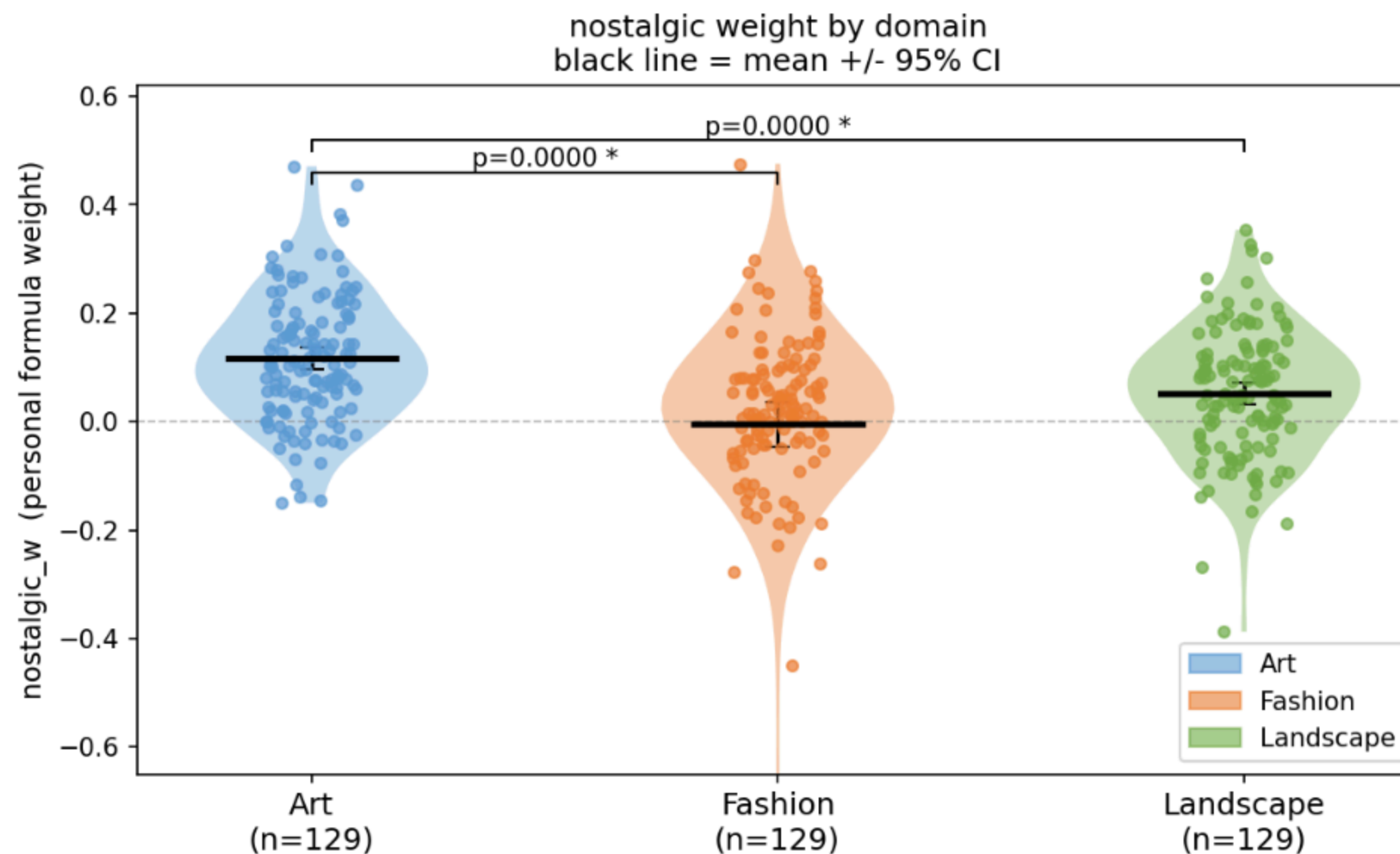
The high baseline (~0.65) reflects shared human preferences, such as almost everyone weighting distasteful negatively. The +0.12 gap on top confirms an identifiable personal signature.



Key Conclusion

A user's beauty formula represents a stable personal signature that does not shift across image types, supporting the personalization approach in our pipeline.

Nostalgia Weight Differs by Domain.



Mean weights

Art mean = +0.12 vs Fashion mean = -0.005

Significance

Mann-Whitney U test: $p < 0.0001$ (art > fashion)

Takeaway

Nostalgic weight is the least cross-domain-stable emotion ($r = 0.138$)

What RQ3 and RQ4 tell us **Together.**

RQ4 confirmed the formula is real

The personal formula is a genuine characteristic of each person and stays stable across contexts.

RQ3 showed it is not just personality

The formula does not reduce to Big Five personality traits in any simple way, and no correlations survived multiple testing.

Experiment 2 explains why

88% of personalization came from how people perceive emotions in the same image, not how they weight them. Personality surveys fail to capture this subjective perception.

① **Personal aesthetic tends to relate with deeper personal metadata**

Personal aesthetic tends to relate more strongly with deeper personal metadata, such as **life experiences**, **cultural background**, or **visual exposure history**. This richer context is more likely to explain variance in subjective emotional perception and support more personalized aesthetic models.

What's next **Two follow-ups.**

1. Run ICI baseline on our split

Read the code from Sia-san and run ICI on the same 5-fold split we used so the comparison with our hybrid model is completely fair.

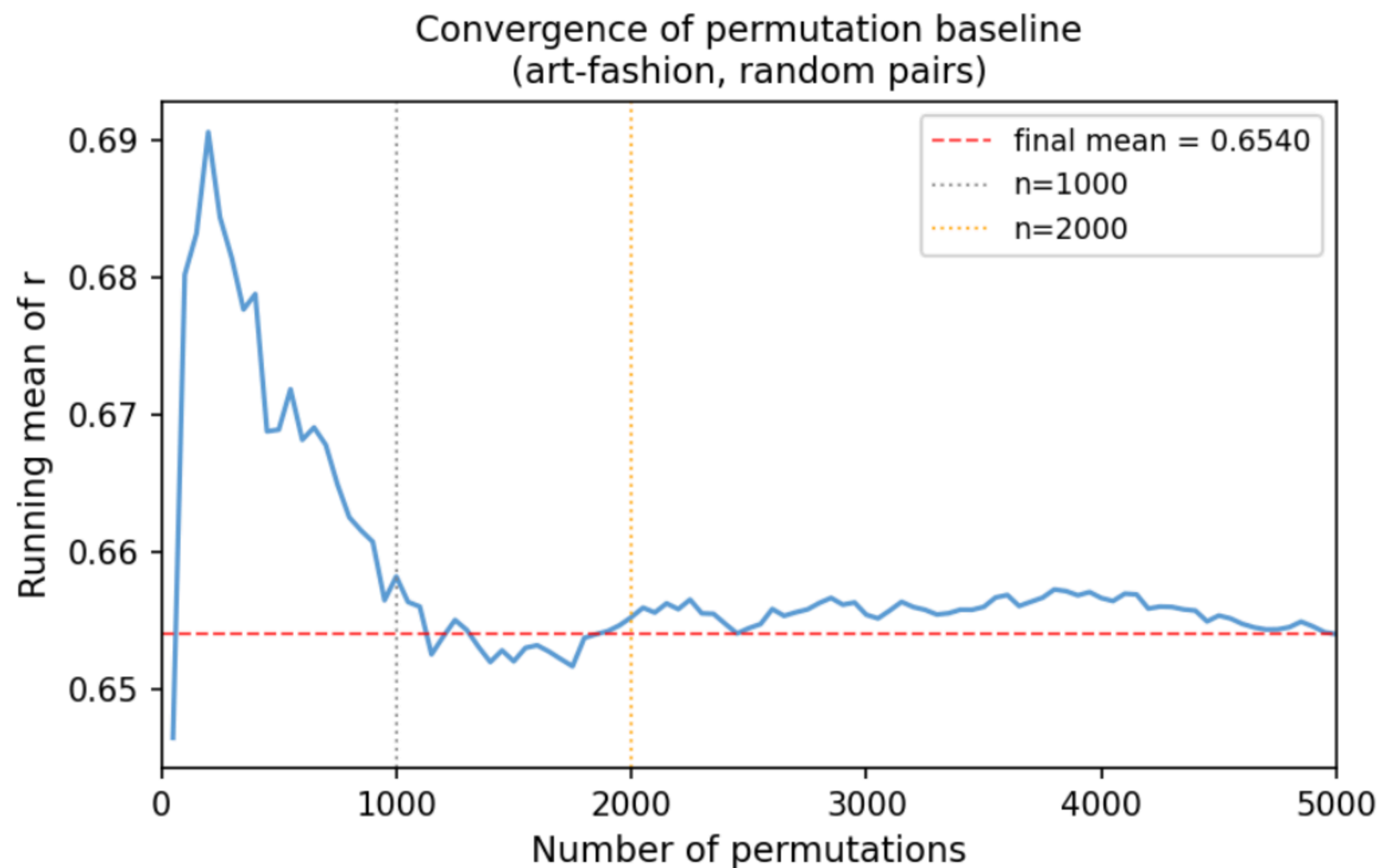
2. Try a stronger image feature

Replace CLIP with hidden representations from a vision-language model. Our current emotion prediction accuracy is $r = 0.27$, which is the main bottleneck in the pipeline.

Thank You!

Happy to take any questions or suggestions.

Permutation test convergence Appendix.



Running mean stabilizes by $\sim n=500$ rounds. Difference between $n=2,000$ and $n=5,000$ is only 0.001, confirming 2,000 rounds is sufficient.

How we measure the relationship

Spearman rank correlation.

What it asks

When one value goes up, does the other tend to go up too?

How it works

1. Convert values to ranks instead of raw numbers.
2. Compute Pearson correlation on the ranked values.

Formula

$$r = 1 - (6 \times \Sigma d^2) / (n \times (n^2 - 1))$$

r ranges from -1 to +1

Close to +1 means the weight is high and the trait score also tends to be high.

Close to 0 means no relationship.

Close to -1 means the weight is high and the trait score tends to be low.

Why Spearman, not Pearson?

Personality survey data may not be normally distributed. Spearman works on ranks, so it is more robust to outliers and does not assume a specific distribution shape.

p-value

It is the probability of seeing this r by chance if there is truly no relationship. Small p means the relationship is less likely to be random.

Why we apply Bonferroni correction.

The multiple testing problem

If we flip a fair coin 35 times, we expect to get an unusual streak at least once just by chance. Statistics behaves the same way.

Testing 35 pairs means we expect about 1–2 results to look significant at $p < 0.05$ even if nothing is real.

Expected false positives

$$35 \times 0.05 = 1.75 \text{ pairs}$$

Bonferroni correction

Divide the significance threshold by the number of tests.

$$\text{New threshold} = 0.05 \div 35 = 0.00143.$$

Result for RQ3

All 3 pairs have p-values between 0.024 and 0.041, which are above 0.00143, so none are confirmed after correction.

Bottom line

We found 3 pairs, which is not much more than the number expected by chance alone.